

Fall 2022 Geometry & Analysis



UCSC MATHEMATICS DEPARTMENT

Jared Marx-Kuo
Stanford University

Variations of Renormalized Volume for Minimal Submanifolds of Poincare-Einstein Spaces

In this talk, we discuss the regularity and renormalized volume of minimal submanifolds, Y , of arbitrary codimension in Poincare-Einstein manifolds, M . Using geometric and microlocal techniques, we derive polyhomogeneous expansions for the minimal submanifold and variations along it. We then present formulae for the first and second variations of renormalized volume. We end with an orthogonality relationship between coefficients in the expansion of Y , for the specific case of $M = H^{n+1}$. In all theorems, we emphasize the Dirichlet-to-Neumann type map and its presence in formulae.



Come join us!

"Tea"
(informal
meet-and-
greet)
before talk

Wednesday
Nov 9

3 PM PST

IN PERSON !

McHenry 4130

Refreshments
in the tea room (McH 4161)
~30m before the talk

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